



Dominik Stiftinger-Lang

Senior Backend & AI Systems Engineer
Go, Python, Machine Learning, AI Agents

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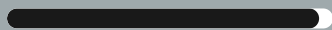
:: Profile summary

Senior engineer, mathematician and founder with nine years in digital-asset and financial systems. Built trading, ML and distributed Go systems, published neural-network research, and led companies, products and projects.

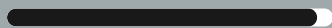
Contracting via ChainBrain FlexCo
Remote from Graz, Austria

:: Expertise

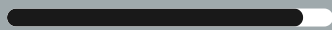
Go & Distributed Systems



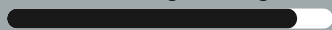
Digital Asset Trading & Risk



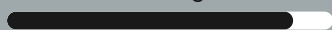
PostgreSQL & APIs



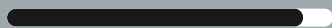
AI-Assisted Engineering



Machine Learning & RL



Company & Project Management



:: Engagement

Available

1 September 2026

Capacity

30-40 hours per week

Standard remote rate

EUR 120/hour excluding VAT

Contract model

B2B via Austrian FlexCo

:: Languages

German Native

English Fluent

:: Selected Projects

2026 – present
ChainBrain FlexCo

AI-Assisted Software Delivery and Quality Control

Selected ChainBrain initiative

- Claude Code and Codex workflows, multi-agent orchestration, reusable skills, and MCP-based tools.
- Evidence-based branch verdicts for AI-generated changes in a financially sensitive Go monorepo.

2022 – present
ChainBrain FlexCo

Distributed Portfolio Management and Trading Platform

Co-Founder, CEO, Lead Backend Engineer and Architect

- Product ownership and Scrum-based delivery alongside hands-on architecture and implementation.
- Production Go platform with Temporal workflows, PostgreSQL, gRPC, OpenAPI, and live client capital.
- Architecture spanning portfolio construction, best execution, reconciliation, reporting, and regulated boundaries.

2020 – present
Nerox / TU Graz /
University of Graz

Production ML and Reinforcement Learning Research

Quantitative Researcher / ML Engineer

- Built the 2020-2021 Apex portfolio ML pipeline: data and feature engineering, TensorFlow/Keras training, automated tuning and backtesting, and staged live trading.
- Published spiking-neural-network research in 2021 and continued reinforcement-learning research for DeFi portfolio constraints from 2022.

:: Technical Focus

- Go and Python
- Distributed Systems
- Best Execution Algorithms
- PostgreSQL and Temporal
- TensorFlow, Keras and RL
- Product Ownership and Scrum
- Claude Code and Codex
- Digital Asset Trading
- MCP and AI Agents
- REST, gRPC and OpenAPI
- Company and Project Management



:: Core Technologies

Languages & Machine Learning

Go / Python
SQL
TensorFlow / Keras
Machine learning / Reinforcement learning

Trading & Architecture

Digital assets
Best execution algorithms
Temporal workflows
REST, gRPC, OpenAPI

Management & Delivery

Company leadership
Project planning
Product ownership
Scrum

:: Project History

ChainBrain FlexCo, Graz

2026 - present

Selected Initiative - AI-Assisted Engineering and Quality Control

- Designed multi-agent workflows with bounded roles for implementation, codebase research, test generation, security and financial-flow audits, and independent verification.
- Standardized Claude Code and OpenAI Codex through reusable skills, task specifications, ownership rules, structured handoffs, and persistent engineering knowledge.
- Integrated MCP-based tools for knowledge retrieval and operational workflows, including a Python knowledge service over structured project documentation.
- Designed an evidence-based QC harness with static and unit checks, debt and reuse analysis, traceable findings, and explicit branch verdicts.

Technologies: Claude Code, OpenAI Codex, MCP, Go, Python, Bash, GitLab CI, Linux

ChainBrain FlexCo, Graz

2022 - present

Co-Founder, CEO, Lead Backend Engineer and Architect

- Established Scrum-based product and engineering delivery across Nerox and ChainBrain, serving as Scrum Master until 2023 and as Product Owner throughout; after the roles were separated, focused fully on Product Ownership.
- Wrote the project plan for an FFG-funded R&D project and managed delivery through final reporting.
- Architected and implemented distributed Go services with Temporal workflows, PostgreSQL persistence, gRPC and OpenAPI interfaces, and idempotent financial processing.
- Designed service and data boundaries for portfolio construction, account state, best execution, reconciliation, reporting, and client-facing APIs.
- Implemented Docker and GitLab CI delivery, Vault and Consul infrastructure, monitoring, incident response, and controlled secrets handling.
- Defined a MiCA-aware architecture with explicit custody, authorization, customer-controlled execution, hosted analytics, and audit boundaries.

Technologies: Go, Python, PostgreSQL, Temporal, gRPC, OpenAPI, REST, Hasura, Docker, GitLab CI, Vault, Consul

ChainBrain FlexCo

2024 - 2026

Selected Project - Multi-Venue Order Management and Execution

- Within ChainBrain, led REST and WebSocket integrations for Binance, Kraken, Bitfinex, Poloniex, and OKX, including venue-specific order semantics and recovery behaviour.
- Developed best-execution algorithms, order-lifecycle state machines, fill and balance reconciliation, exposure controls, a B-tree orderbook, and execution-quality measurement.
- Operated production spot and futures rebalancing with sub-1 bps median implementation shortfall and no exposure-limit breaches.

Technologies: Go, PostgreSQL, Temporal, REST, WebSocket, gRPC, Docker, best execution, implementation shortfall



:: Earlier Experience

Nerox GmbH, Graz

2019 – Aug 2026

Co-Founder, Head of Quantitative Finance, later Chief Risk Officer

- Co-designed and built Apex (2020–2021), an autonomous portfolio-management pipeline spanning real-time Go/WebSocket market-data collection, PostgreSQL storage, model deployment, and live trading.
- Implemented R/SQL/Python feature engineering and TensorFlow/Keras training with a ResNet-derived allocation model, portfolio-value objective, automated hyperparameter tuning, validation, backtesting, and staged capital rollout.
- Designed and operated HFT, arbitrage, trend-following, grid-trading, and DeFi strategies, with responsibility for risk controls, validation, deployment, and production performance.
- Owned quantitative design, validation, deployment, and communication with business stakeholders.
- Established and facilitated Scrum-based delivery across software and quantitative work, coordinating roadmap priorities, sprint delivery, and architecture decisions.

Technologies: Python, TensorFlow, Keras, R, PostgreSQL, SQL, Go, WebSocket, HFT, arbitrage, trend following, grid trading, DeFi, machine learning, reinforcement learning

Nerox GmbH, Graz

2017 – 2019

Software Developer

- Implemented HFT, arbitrage, trend-following, and grid-trading strategies plus low-latency Go exchange adapters using REST and WebSocket market-data and order APIs.

Technologies: Go, REST, WebSocket, SQL, Linux, algorithmic trading, HFT, arbitrage, trend following, grid trading, low-latency systems

Graz University of Technology

2020 – 2021

Research Assistant – Spiking Neural Networks

- Developed and evaluated spiking-neural-network models in computational neuroscience; co-authored the resulting 2021 publication with Christoph Stoeckl and Wolfgang Maass.

Technologies: Spiking neural networks, recurrent neural networks, PyBullet, evolutionary strategies, computational neuroscience, high-performance computing

:: Education

In progress

University of Graz

PhD Mathematics

Portfolio optimization under DeFi staking and lock-up constraints. Developed reinforcement-learning experiments from 2022, including a Gym environment and a TensorFlow/Keras actor-critic implementation.

2017 – 2019

Graz University of Technology

MSc Mathematics

Numerical methods for partial differential equations

2013 – 2017

Graz University of Technology

BSc Mathematics, Technomathematics

:: Selected Publications

2025

A Multi-Factor Investment Framework for Crypto Asset Management

Crypto Valley Association Research Journal. Highest reviewer rating among 12 selected papers.

2021

Structure induces computational function in networks with diverse types of spiking neurons

Stoeckl, Lang, Maass. bioRxiv 10.1101/2021.05.18.444689.

2019

A Study of Boundary Element Methods for Electrostatic Transmission Problems

MSc thesis, Graz University of Technology.

